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Title of the article: *Isolation and characterization of human salivary gland cells for stem cell transplantation to reduce radiation-induced hyposalivation*

The content extracted is as follows:

Radiation-induced sterilization of the primitive glandular stem cells which prevents the replenishment of saliva-producing cells seems to be the main cause for irreversible late hyposalivation [4].

Transplantation of salivary gland stem cells potentially offers a therapy to rescue irradiated salivary glands.

For many tissues experimental forms of stem cell transplantation have been shown to result in improvement of tissue function, such as that for bone marrow, skin [2,12], mammary gland [13,15], eye [14], kidney [4], brain [10], and heart [1].

However, only hematopoietic stem cell transplantation and epithelial stem cell-based treatments for skin burns and corneal disorders are routinely applied clinically.

The first attempt to isolate and transplant rat salivary gland stem/progenitor cells was described in 2004, but failed to show formation of saliva-producing acinar cells or functional improvement [16].

Later, stem cells that were selected using the stem cell markers Sca-1, c-Kit or $\alpha^6\beta^1$ integrin cells from, respectively, mouse [5,11] and rat [6] glands were shown to contribute to tissue regeneration when transplanted via the portal vein into partially hepatectomized animals.

The first success in developing an in vitro stem cell assay based on colony forming units was performed in 2006 when Kishi and colleagues developed a clonal colony assay for salivary gland cells [6].

However, due to the lack of in vivo functional assessment, these as such determined colony forming cells could not be defined as stem cells [6].

Recently, however, in a series of studies [3,7,9], we showed that salivary glands do contain cells that when stimulated appropriately have the capacity to regenerate damaged tissue after irradiation.

Moreover, we developed a culture and isolation procedure to obtain a c-Kit⁺ cell population from mice submandibular glands that contain stem cells which when transplanted in irradiated salivary gland, prominently restored function and morphology [8].

The aim of this study was to expand our findings to human salivary glands for future development of clinical application.

Since this therapy might be applicable to older patients, we also investigated the relationship between age and the number of salivary gland stem cells.

Materials and methods

Irradiated human salivary gland tissue was obtained from a patient (after informed consent) with a squamous cell carcinoma of the oral cavity in whom a neck dissection procedure was performed. This patient was operated and irradiated 15 years ago for an oropharyngeal tumor T4N1M0.

Per gland, between 2 and 4.6 $\times 10^6$ cells were isolated. The culture medium consisted of DMEM/F12 (Invitrogen, Carlsbad, CA; 11320 074), penicillin, streptomycin, glutamax, EGF (20 ng/mL), FGF-2 (20 ng/mL), N-2 (serum-free supplement for culturing neural cells) (1/100), insulin (10 μ g/mL) and dexamethasone (1 nM) further depicted as salivary gland medium (SGM).

Human parotid and submandibular salivary glands were obtained from patients (after informed consent) with a squamous cell carcinoma of the oral cavity in whom a neck dissection procedure was performed. None of the patients had received any other cancer treatment before surgical procedure.

Results

The successful transplantation of a murine c-kit⁺ salivary gland stem cell containing cell population that restored the function of irradiated glands [8], warrants translation of the results to humans.

The present study was performed to compare some basics in radiation response and stem cell characteristics between human and mouse salivary glands.

The success of the mouse salivary gland stem cell transplantation is partly due to the typical response of the tissue to radiation with disappearing acinar cells but with a ductal compartment remaining largely intact.

Since, the stem cells of the salivary gland reside in the ducts [8], they provide the environment for engraftment of the transplanted cells.

Therefore, we first studied the histopathology of irradiated human salivary gland.

Normal human salivary parenchyma comprises of acini, which similar to mice are composed of triangle-shaped saliva-producing acinar cells, and a branchy duct system (Fig. 1A, C, and E).

After radiotherapy, acinar cells are replaced by brotic tissue but similar to mice still some ductal tissue remains (Fig. 1B, D, and F), which may allow for the engraftment of stem cells in patients.

Next, we investigated the presence of salivary gland stem cells in human salivary gland, using the same method of culturing salispheres as in our mice studies [8].

This assay reveals long-term self-renewal ability in vitro to suggest the presence of a stem cell population in human salispheres.

Recently, we showed that especially c-Kit⁺ cell isolated from 3day-old spheres cultured from mouse salivary glands contained cells that were able to transdifferentiate in all salivary gland cell types, to regenerate irradiated glands, and to self-renew in vivo [8].

Therefore, we investigated human salivary glands for the presence of c-Kit⁺ cells.

Using immuno- uorescentstaining,normalhuman salivary gland clearly showed c-Kit⁺ cells in excretory ducts of both submandibular and parotid glands (Fig. 5A and B, respectively).

However, patients that are treated for head and neck cancer are often relatively much older and potentially may have different numbers of stem cells.

Since we showed that the number of salispheres that can be grown from mouse salivary glands represent the number of stem cells present [9], we testedfor a possible relationship between age and the number of salivarygland stem cells.To this end,we quantitatively compared thesalispherecultureofold miceand young mice.Salispheres were cultured from 10 old mice and the number of salispheresthat couldbe grownwas comparedtoyoungmice.After3 days of culturing, salisphereswerecountedandsubsequently dispersedinto a cell suspension, put into matrigel and cultured for another 10 days.

Discussion

Recently, we discovered a population of c-Kit-expressing cellsin mice with remarkable capability to restore salivary glands damaged by radiation [8].

Salispheres cultured from rodent submandibular glands contained cells that expressed many stem cell markers (e.g., Sca-1, c-Kit, Musashi-1) and were able to differentiate into all salivary gland lineages.

After stem cell enrichment by ow cytometric selection using c-Kit as a single marker, as few as 100 c-Kit⁺ cells were able to completely restore salivary gland function and morphology 3 months after transplantation.

This study was the rst evidence for the potential use of stem cell transplantation to functionally rescue salivary gland damaged by irradiation.

In this study, we investigated the presence and in vitro potential of human salivary gland stem cells.

We determined the characteristics of human salivary gland for the following aspects: morphology and pathology, salisphere formation, in vitro differentiation, self-renewal ability and the presence of c-Kit stem cell marker.

Although human and mouse salivary glands are not exactly the same, the tissue architecture after irradiation looks remarkably similar.

In both species, the ductal compartment necessary for stem cell engraftment largely remains intact.

Moreover, salisphere formation of human salivary gland cells was very similar to mice.

These results indicate that human salispheres do contain cells with stem cell-like properties.

The potential existence of stem cells that may be used for transplantation purposes is further substantiated by the location of c-Kit⁺ cells in the excretory ducts of both human submandibular and parotid glands.

This may be due toa lower stem cell number in older people such asthe patients with head and neck cancer used in this study.

Therefore, we conclude that it may very well be possible to develop human adult stem cell therapy to restore the function of irradiated salivary glands in patients.

And, can we get suf cient and potent enough stem cells to establish long term regeneration of salivary glands or do we have to expand human salivary gland stem cells in vitro before transplantation?

Recently, we have shown that in vivo expansion of salivary gland stem cells is attempted [9] and it may thus also be possible in vitro.

Human and mouse salivary glands contain a similar putative stem cell population, which might have

potential to reduce radiotherapy-induced salivary gland dysfunction in the future.

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